17. Complete enumeration of a precisely defined molecular-graph class

Pure Thought research portfolio

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Revised specification — 2026-09-11. Detailed critique · Original description · Evidence standard

Plain-language guide

The problem in everyday terms

A molecular formula tells us how many atoms of each kind are present, but not how they are connected. The same counts can describe different structures: butane and isobutane are a familiar example. This problem asks for every distinct connection pattern allowed by an explicit set of bonding rules, with a proof that none was counted twice and none was missed.

Key terms

- **Molecular formula** — A count of atoms by element, such as C_4H_{10}.
- **Constitutional isomer** — A structure with the same molecular formula as another but a different pattern of atom connections.
- **Molecular graph** — A mathematical representation with atoms as vertices and bonds as edges carrying bond-order labels.
- **Valence grammar** — The exact bonding rules permitted in this enumeration; it deliberately defines a restricted chemical model.
- **Bond order** — The bond label distinguishing single, double and triple bonds in this graph model.
- **Graph isomorphism** — A relabeling of atoms that preserves element types and bonds, showing two drawings represent the same graph.
- **Canonical labeling** — A consistent representative label for a graph, used to identify duplicates.
- **Stereoisomer** — A distinction in spatial arrangement not captured merely by the basic connection graph.
- **Completeness certificate** — Evidence that the enumeration covers every object in the specified graph class.

Why this matters

Complete enumeration turns an intuitive picture of "possible structures" into a checkable mathematical set. Such sets are useful for testing structure-generation methods and exploring how bonding constraints shape chemical possibilities. The guarantee is especially valuable when a search is used to support a claim that no alternative structure exists.

What progress would mean

Success would supply exact graph counts and explicit structures under the chosen rules. A graph's presence in the list would not imply that the corresponding molecule is stable, easily synthesized or realizable in a particular three-dimensional shape.

Core question

For a fixed formula and explicit valence grammar, can every connected constitutional graph be generated exactly once, with a checkable completeness argument?

Scope and assumptions

Initially allow neutral closed-shell C,H,O graphs with bond orders 1,2,3, no self-bonds, and valences 4,1,2. Hydrogens may be implicit if reconstructed uniquely from residual valence. Specify whether different Kekulé graphs are distinct; the baseline uses literal bond-order graphs and no aromatic equivalence.

Constitutional graph validity is not chemical stability or synthesizability. Stereoisomers, conformers, ions, radicals, isotopes, and 3D geometries require added definitions and are not baseline outputs.

Mathematical target

For labeled heavy atoms set symmetric b_ij∈{0,1,2,3}, b_ii=0, h_i=v_i−Σ_j b_ij≥0, and Σ_i h_i=n_H. Require connectivity and the requested counts of element labels. Quotient by permutations preserving element labels and bond orders.

Prove completeness via canonical augmentation with a proved parent rule, or exhaustive labeled enumeration plus verified orbit reduction. A SAT route needs the exact encoding and orbit-blocking/symmetry-breaking justification; the final UNSAT proof must cover all remaining admissible graphs.

Required outputs and proof obligations

Export canonical graphs, reconstructed hydrogens, counts, labeling conventions, and generation/coverage proof. Provide a validator independent of the optimized generator. If a resource limit stops enumeration, output a partial catalog and coverage status rather than "all isomers."

Validation and rejection controls

Recover alkane constitutional counts for C_4H_{10} and C_5H_{12}, namely 2 and 3. Cross-check all tiny instances with brute force. Random relabelings must preserve canonical codes. Verify each graph's formula and weighted valence; reject disconnected mixtures.

Milestones and research extension

Baseline: complete small-formula catalog with independent counts. Strong: complete $C_6H_{12}O$ under the stated grammar. Research extension: stereochemical orbit enumeration with meso cases and explicit stereocenter/double-bond conventions; keep conformational continua separate.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

Primary references and their role

- McKay, Isomorph-free exhaustive generation — Canonical generation and completeness methodology.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

Displayed formulation and claim boundary

Canonical specification.

Let $\mathcal{G}$ contain exactly the graphs satisfying the fixed atom, bond, valence and connectivity grammar. Then

$$\text{catalog} = \mathcal{G}/\cong .$$

Isomorphisms preserve the declared atom and bond labels. Every output must satisfy the grammar, no two representatives may be equivalent, and every allowed input must map to a representative. Canonical labeling proves deduplication only; completeness needs a coverage proof for generation and pruning. Stereochemistry requires a separately declared equivalence model.

Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

Detailed findings

1. **Major — chemical universe (39–47).** Fixed valences do not describe every molecule of a formula. Charge, radicals, aromatic equivalence, stereochemistry and disconnected structures change the answer. The grammar now defines what "all" means.
2. **Major — canonicity is not completeness (96–99).** No duplicate outputs does not prove no missing outputs. A coverage argument must audit pruning and symmetry breaking.

3. **Major — Pólya shortcut (101–108).** Burnside/Pólya counting requires a specified group action and correct fixed-point counts under the graph constraints. Writing a cycle-index sum alone does not enumerate connected valence-constrained graphs.
4. **Major — physical and spatial claims.** SMILES and a valence-correct graph do not certify a stable 3D structure. The rewrite preserves an exact combinatorial task and treats stereochemistry as a separately verified extension.

What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For a fixed formula and explicit valence grammar, can every connected constitutional graph be generated exactly once, with a checkable completeness argument?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

Acceptance criteria to use instead

Baseline: complete small-formula catalog with independent counts. Strong: complete $C_6H_{12}O$ under the stated grammar. Research extension: stereochemical orbit enumeration with meso cases and explicit stereocenter/double-bond conventions; keep conformational continua separate.

Sources supporting the corrections

- McKay, Isomorph-free exhaustive generation — Canonical generation and completeness methodology.

Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

Four evidence levels

Label	What it establishes	What it does not establish
Numerical evidence	A reproducible computation at stated precision, sample size, and discretization	Exact truth, continuum coverage, or a universal theorem
Finite-model certificate	An exact/validated statement about the specified finite model or finite relaxation	An omitted tail, thermodynamic limit, or physical realization
Conditional mathematical theorem	A proof over the stated domain with every hypothesis and error controlled	Truth outside those assumptions or a demonstrated physical system
Formal proof	A checked theorem in a named proof assistant, with imported axioms/dependencies listed	Correct physical modeling merely by virtue of formalization

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means "not excluded"; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.

- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

Checking specific certificate types

LP/SDP/SOS: verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

Interval numerics: state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

SAT/SMT: a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

Statistics: specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.